

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

MI-7001-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Advanced Surveying

Time : Three Hours

Maximum Marks : 70

- Note:** i) Answer any five questions.
ii) All questions carry equal marks.

1. a) From an eccentric station S, 12.25 m to the west of the main station B, the following angles were measured.
angle BSC = $76^{\circ} 25' 32''$; angle CSA = $54^{\circ} 32' 20''$
The stations S and C are to the opposite sides of the line AB, Calculate the correct angle ABC if the lengths AB and BC are 5286.5 m and 4932.2m respectively.
 - b) The altitude of two proposed stations A and B 130 km apart are respectively 220 m and 1160 m. The altitudes of two points C and D on the profiles between them are respectively 308 m and 632 m, the distances being AC = 50 km and AD = 90 km. Determine whether A and B are intervisible, and if necessary, find the minimum height of a scaffolding at B, assuming A as the ground station.
2. State the factors affecting selection sites of triangulation stations.

MI-7001-CBGS

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[2]

3. Explain the different methods of stop surveying.
4.
 - a) Discuss any two methods of azimuth determination of a line by astronomical observation.
 - b) At a place (latitude 35° N, longitude $15^{\circ} 30'$ E), the following observations were taken on a star:
Clockwise angle between reference mark and star = $36^{\circ} 28' 18''$
R.A. of the star = 10 h 12 m 6.3 s
Declination of the star = $20^{\circ} 06' 48''$. 4
G.M.T. of the observation = 19 h 12 m 28.6 s
G.S.T. of G.M.M. = 10 h 12 m 36.2 s
Determine the true bearing of the line.
5.
 - a) Write a note on ground control for aerial photogrammetry.
 - b) An area 30 km long in the north-south direction and 24 km in the east-west direction is to be photographed with a lens having 300 mm focal length. The photograph size is $20 \text{ cm} \times 20 \text{ cm}$. The average scale is to be 1: 12000 effective at an elevation of 400 m above datum. Overlap is to be at least 60% and the side lap is to be at least 30%. An intervalometer will be used to control the interval between exposures. The ground speed of the aircraft will be maintained at 200 km per hour. The flight lines are to be laid out in a north-south direction on an existing map having a scale 1: 60000. The two outer flight lines are to coincide with the east and west boundaries of the area. Determine the data for the flight plan.

[3]

6.
 - a) Discuss the constructional features and use of Gyrotheodolite.
 - b) Discuss the constructional features and use of Phototheodolite.
7.
 - a) Explain the working principle and use of GPS in mining industry.
 - b) Explain the working principle of total station with neat sketch.
8. Write short notes on : (any two)
 - a) Geodimeter
 - b) Electronic distance meter
 - c) Check base lines
